BEING QUASI-COMPACT IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's closing phrase 'and we win' is quoted verbatim.

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Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Base-change stability and Zariski locality of quasi-compactness. Schemes, Lemmas schemes-lemma-quasi-compact-preserved-base-change and schemes-lemma-quasi-compact-affine, invoked at proof lines 5694-5697. Supplies the two preliminary halves of fpqc locality.
- The general principle for a property that is fpqc local on the base. Same chapter, Lemma lemma-descending-properties-morphisms, invoked at proof line 5704. Concludes the locality statement.

Source reference. Original lemma 02KP: sources/stacks-descent/pinned-original-source.tex, source lines 5624–5624. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01K4: sources/stacks-chow/tags.txt, source lines 2037–2037. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01K5: sources/stacks-chow/tags.txt, source lines 2038–2038. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is quasi-compact" is fpqc local on the base.*

Proof. A base change of a quasi-compact morphism is quasi-compact, see Schemes, Lemma 01K5. Being quasi-compact is Zariski local on the base, see Schemes, Lemma 01K4. Finally, let $S' \to S$ be a flat surjective morphism of affine schemes, and let $f : X \to S$ be a morphism. Assume that the base change $f' : X' \to S'$ is quasi-compact. Then X' is quasi-compact, and $X' \to X$ is surjective. Hence X

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is quasi-compact. This implies that f is quasi-compact. Therefore Lemma 02KP applies and we win. $\square$